The American Journal of Medicine (2007) 120, 901-910

THE AMERICAN JOURNAL of MEDICINE®

CLINICAL RESEARCH STUDY

Prevention of Central Venous Catheter-Associated Thrombosis: A Meta-analysis

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ABSTRACT

PURPOSE: Anticoagulant prophylaxis in patients with central venous catheters is controversial. We performed a meta-analysis of randomized controlled trials of anticoagulant prophylaxis in patients with central venous catheters.

METHODS: MEDLINE and EMBASE were searched up to May 2006, supplemented by manual searches of conference proceedings and bibliographies.

RESULTS: Fifteen trials were included. Unfractionated heparin infusion, oral fixed low-dose vitamin K antagonist, and subcutaneous low-molecular-weight heparin were evaluated. For all catheter-associated deep vein thrombosis (symptomatic and asymptomatic combined), the summary relative risks ranged from 0.31 to 0.73 (all achieved statistical significance). For symptomatic deep vein thrombosis, the summary relative risks ranged from 0.28 to 0.72, but did not achieve statistical significance for any individual regimen.

CONCLUSION: Anticoagulant prophylaxis is effective for preventing all catheter-associated deep vein thrombosis in patients with central venous catheters. The effectiveness for preventing symptomatic venous thromboembolism, including pulmonary embolism, remains uncertain. © 2007 Elsevier Inc. All rights reserved.

KEYWORDS: Anticoagulant; Catheter; Meta-analysis; Prophylaxis; Thrombosis; Venous

Central venous catheters are increasingly used in clinical practice, with a 27% to 67% incidence of catheter-associated deep vein thrombosis,[1-3] which is most often asymptomatic. Pulmonary embolism occurs in 15% to 36% of patients with symptomatic catheter-associated deep vein thrombosis.[2,3] Thus, preventive measures for catheter-associated venous thromboembolism might be warranted.

Although studies evaluating thromboembolism prophylaxis in patients with central venous catheters have been published, some report conflicting results, and evidence-based recommendations for clinical practice have varied.[4,5] We performed a meta-analysis to clarify the efficacy and safety of anticoagulant administration for prevention of catheter-associated venous thromboembolism.

METHODS

We searched MEDLINE (1964 to 2006), EMBASE, and conference proceedings from The American Society of Hematology (2002 to 2005), The American Society of Clinical Oncology (1999 to 2006), and the International Society of Thrombosis and Haemostasis (2001 to 2005) using the following terms: "Thromb:.mp or hypercoag:.mp or occlu:.mp"; "anticoag:.mp or anti-vitamin k.mp or thromb: prophylax:.mp or heparin:.mp"; "catheter:.mp or cannul:.mp or central lin:.mp or port:.mp or central venous:.mp or vascular access:.mp"; as well as generic and trade names for anticoagulants and catheters. Limits were set for human subjects and English language. Authors were contacted for additional information if required. Bibliographies were screened for relevant articles.

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doi:10.1016/j.amjmed.2007.05.010

Objective criteria for study inclusion were defined a priori. Included were studies that: enrolled adult patients, evaluated systemically administered anticoagulant prophylaxis other than flushing of catheters with heparin or saline, evaluated prophylaxis in patients with central-venous catheters or peripherally inserted central catheters, objectively evaluated patients for deep vein thrombosis, and were randomized controlled trials.

Two reviewers (ACK, SWR) evaluated each article using predefined criteria that used established methodological standards for meta-analyses evaluating therapy.[6-9] A third reviewer (TLW) resolved disagreements by blinded adjudication.

The strength and quality of each study was assessed according to the following: random and concealed group allocation; consecutive enrollment; patient spectrum; group similarity at baseline; treatment similarity other than the intervention being studied; blinding of patients, clinicians, and outcomes assessors; a priori definition of major and minor bleeding events; and use of an objective test for the evaluation of deep vein thrombosis.

The frequency of the following outcomes was recorded: all catheter-associated deep vein thrombosis; symptomatic catheter-associated deep vein thrombosis; symptomatic pulmonary embolism; major bleeding; minor bleeding; and death from all causes. Studies were included in the analysis of "all catheter-associated deep vein thrombosis" if patients underwent mandatory screening with an objective test for deep vein thrombosis. The analysis of "symptomatic catheter-associated deep vein thrombosis" included all studies that reported symptomatic events confirmed by an objective diagnostic test.

For purposes of data extraction, clinically overt bleeding was defined as major if it resulted in a decrease in hemoglobin of 2 g per deciliter or more, required the transfusion of 2 or more units of blood, was retroperitoneal, intracranial, intraocular, or contributed to death.[10] Minor bleeding was defined as clinically overt bleeding not meeting criteria for major bleeding.

Statistical Methods

Statistical analyses adhered to established standards for meta-analysis.[11-13] We assessed the validity of combining results from individual studies using the Q statistic to assess for heterogeneity.[13] We considered a P value of .10 or less statistically significant evidence of heterogeneity, which, if present, would imply that the included studies estimate different population effect sizes.[13] Summary relative risks for each outcome were generated using the fixed effects model, which assumes that all studies estimate the same population effect size. Analyses were repeated using the more conservative random effects model, which assumes the studies estimate different population effect sizes.[13] Statistical significance of summary treatment effects was determined using a Z test, with a P value $\leq .05$ considered statistically significant.[13]

Sensitivity analyses were performed to evaluate the robustness of our results after removing individual studies from the analysis. We repeated the analyses, limiting the included studies to those that were double-blind, and those published after the year 2000. Funnel plots were generated to determine whether unpublished studies with nonsignificant results were underrepresented in our analysis, leading to publication bias that may yield overly optimistic summary relative risks.[13] Calculations were performed using Comprehensive Meta-Analysis (Biostat, Inc, Englewood, NJ).

CLINICAL SIGNIFICANCE

- In patients with central venous catheters, anticoagulant prophylaxis reduces risk of all (combined symptomatic and asymptomatic) catheter-associated deep vein thrombosis.
- Effective prophylactic treatments include low-dose unfractionated heparin infusion, fixed low-dose vitamin-K antagonists, and prophylactic doses of low-molecular weight heparin.
- These treatments do not increase risk of major bleeding.
- Whether these treatments prevent symptomatic deep vein thrombosis, including pulmonary embolism, remains uncertain.

DATA SYNTHESIS

Figure 1 shows the process of study selection. The literature search identified 70 studies.[14-83] Fifty-seven did not meet the predefined eligibility criteria: 24 were reviews,[14-37] 13 were retrospective,[38-50] 2 were opinions,[51,52] 1 was a case series,[53] 9 were nonconcurrently controlled,[54-62] and 8 were not randomized.[63-70] Thirteen were randomized[71-83] and were included.

Our search of conference proceedings yielded 11 abstracts.[84-94] Four were retrospective [84-87] and were excluded. Two randomized trials were excluded [88,89] due to incomplete information. Three were also published as full manuscripts and were included as such.[90-92] Two remaining abstracts of randomized controlled trials were included.[93-94]

Study Characteristics

Table 1 (available online) shows the characteristics of 15 randomized controlled trials included for data extraction.[71-83,93-94] All studies included only upper-extremity central venous catheters. Ten included only cancer patients,[74-76,80-83,93,94] and 5 included only patients receiving total parenteral nutrition.[71-73,78,79] Five different anticoagulant medications were evaluated: low-dose unfractionated heparin infusion, fixed low-dose vitamin-K antagonists (warfarin or acenocoumarin), and subcutaneous low-molec-

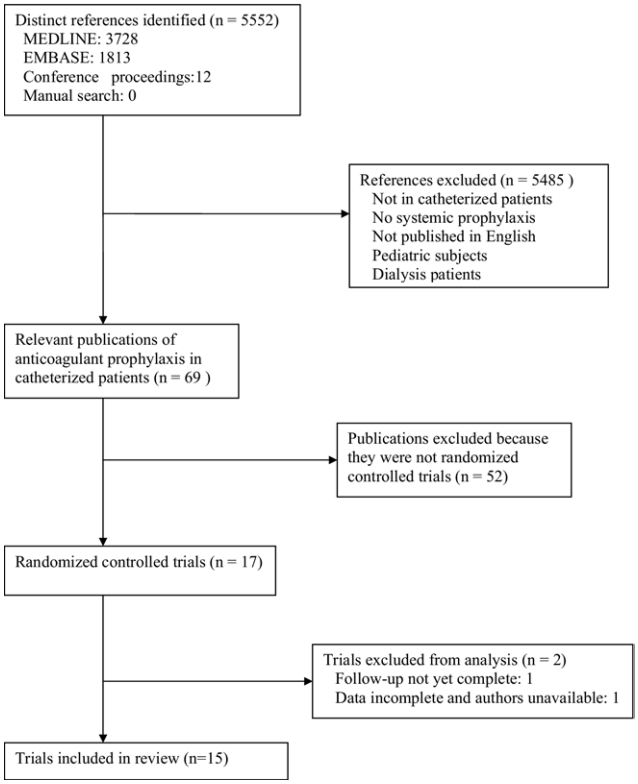

Figure 1 Flow diagram of the trial selection process.

ular-weight heparins (dalteparin, enoxaparin, and nadroparin).[71-74,76-79,82,94] One study compared nadroparin 2850 units daily with warfarin 1 mg daily,[80] and one compared dalteparin 5000 units daily with acenocoumarin 1 mg daily.[94]

Study Quality

Table 2 (available online) summarizes the measures of quality assessed for each study.

All Deep Vein Thrombosis

The results of analysis of all catheter-associated deep vein thrombosis are shown in Table 3 and Figure 2. For low-dose unfractionated heparin infusion,[71-73,77-79] the summary relative risk was 0.31 (95% CI, 0.13 to 0.71). For fixed low-dose vitamin-K antagonist prophylaxis,[74,94] the summary relative risk was 0.37 (95% CI, 0.26 to 0.52), and for low-molecular-weight heparin,[75,81,83,93,94] the summary relative risk was 0.72 (95% CI, 0.57 to 0.90). No analysis demonstrated statistical evidence of heterogeneity. For studies comparing low-molecular-weight heparin to fixed low-dose vitamin-K antagonist[80,94] the summary relative risk was 1.88 for low-molecular-weight heparin (95% CI, 1.28 to 2.75) with no statistical evidence of heterogeneity.

Symptomatic Deep Vein Thrombosis

The results of the analysis for symptomatic catheter-associated deep vein thrombosis are shown in Table 4 and Figure 3. For low-dose unfractionated heparin infusion,[72,77] the summary relative risk was 0.28 (95% CI, 0.05 to 1.69). For fixed low-dose vitamin-K antagonist,[74,76,82] the summary relative risk was 0.60 (95% CI, 0.30 to 1.20). For low-molecular-weight heparin,[75,81,83] the summary relative risk was 0.69 (95% CI, 0.30 to 1.59). The summary relative risk using any anticoagulant prophylaxis[72,74-77,81-83] was 0.59 (95% CI, 0.35 to 0.97). There was no statistical evidence of heterogeneity.

Pulmonary Embolism

Five trials reported documentation of symptomatic pulmonary embolism.[73,75,79,81,82] The summary relative risk using any anticoagulant prophylaxis for preventing symptomatic pulmonary embolism was 1.96 (95% CI, 0.52 to 7.45) with no statistical evidence of heterogeneity.

Bleeding

The results of the analysis for major bleeding are shown in Table 5 and Figure 4. For low-dose unfractionated heparin infusion,[72,77,78] the summary relative risk was 0.74 (95% CI, 0.17 to 3.22). For fixed low-dose vitamin-K antagonist,[76,82] the summary relative risk was 0.24 (95% CI, 0.03 to 2.13). For low-molecular-weight heparin,[75,81,83,93] the summary relative risk was 0.66 (95% CI, 0.12 to 3.68). The summary relative risk of major bleeding with any anticoagulant prophylaxis [72,75-78,81-83,93] was 0.54 (95% CI, 0.20 to 1.42). There was no statistical evidence of heterogeneity.

Seven studies reported minor bleeding events.[72,75,76,78,81-83] For low-dose unfractionated heparin infusion,[72,78] the summary relative risk was 3.74 (95% CI, 0.36 to 38.31). For fixed low-dose vitamin-K antagonist,[76,82] the summary relative risk was 1.76 (95% CI, 0.49 to 6.40). For low-molecular-weight heparin,[75,81,83] the relative risk was 1.32 (95% CI, 0.87 to 2.01). The summary relative risk of minor bleeding with any anticoagulant prophylaxis[72,75,76,78,81-83] was 1.40 (95% CI, 0.94 to 2.07). There was no statistical evidence of heterogeneity.

All-Cause Mortality

Seven trials reported documentation of all-cause mortality.[74-77,81-83] For the single trial using low-dose unfractionated heparin infusion,[77] the relative risk was 0.97 (95% CI, 0.14 to 6.67). For fixed low-dose vitamin-K antagonist,[74,76,82] the summary relative risk was 0.95 (95% CI, 0.62 to 1.46). For low-molecular-weight heparin,[75,81,83] the summary relative risk was 1.57 (95% CI, 0.54 to 4.58). The summary relative risk of mortality using any anticoagulant prophylaxis[74-77,81-83] was 1.03 (95% CI, 0.70 to 1.52). There was no statistical evidence of heterogeneity.

Sensitivity Analyses

For low-dose unfractionated heparin infusion, the summary relative risk for symptomatic catheter-associated deep vein thrombosis in patients receiving parenteral nutrition increased from 0.28 to 1.00 (95% CI, 0.02 to 48.09) after

Table 3 Relative Risk of All Catheter-Associated Deep Vein Thrombosis with Anticoagulant Prophylaxis

Citation	Prophylaxis n/n (%)	No Prophylaxis n/n (%)	Relative Risk	(95% Confidence Interval)	*P* Value
Fixed low-dose heparin					
Abdelkefi et al, 2004[77]	1/65 (1.5)	8/63 (12.7)	0.1212	(0.0156-0.9409)	.0136
Brismar et al, 1982[78]	0/23 (0)	4/26 (15.4)	0.1250	(0.0071-2.2036)	.0804
Fabri et al, 1982[71]	2/24 (8.3)	7/22 (31.8)	0.2619	(0.0608-1.1290)	.0449
Fabri et al, 1984[72]	0/20 (0)	0/20 (0)	1.0000	(0.0208-48.0855)	1.0000
Macoviak et al, 1984[73]	2/17 (11.8)	1/20 (5.0)	2.3529	(0.2331-23.7458)	.4525
Ruggiero et al, 1983[79]	0/17 (0)	0/17 (0)	1.0000	(0.0210-47.7081)	1.0000
Fixed	5/166 (3.0)	20/168 (11.9)	0.3053	(0.1320-0.7061)	.0055
Random	5/166 (3.0)	20/168 (11.9)	0.3450	(0.1357-0.8771)	.0254
Low-molecular-weight heparin					
Conte et al, 2003[93]	4/38 (10.5)	3/20 (15.0)	0.7018	(0.1738-2.8332)	.6191
DeCicco et al, 2006[94]	46/114 (40.4)	63/114 (55.3)	0.7302	(0.5531-0.9638)	.0242
Karthaus et al, 2006[81]	20/294 (6.8)	11/145 (7.6)	0.8967	(0.4416-1.8209)	.7631
Monreal et al, 1996[75]	1/16 (6.3)	8/13 (61.5)	0.1016	(0.0145-0.7108)	.0014
Verso et al, 2005[83]	22/155 (14.2)	28/155 (18.1)	0.7857	(0.4708-1.3112)	.3542
Fixed	93/617 (15.1)	113/447 (25.3)	0.7162	(0.5689-0.9018)	.0045
Random	93/617 (15.1)	113/447 (25.3)	0.7345	(0.5645-0.9557)	.0216
Fixed low-dose vitamin K antagonist					
Bern et al, 1990[74]	4/42 (9.5)	15/40 (37.5)	0.2540	(0.0921-0.7004)	.0027
DeCicco et al, 2006[94]	26/120 (21.7)	63/114 (55.3)	0.3921	(0.2686-0.5723)	<.0001
Fixed	30/162 (18.5)	78/154 (50.6)	0.3655	(0.2561-0.5217)	<.0001
Random	30/162 (18.5)	78/154 (50.6)	0.3718	(0.2609-0.5300)	<.0001
Fixed combined (13)	128/945 (13.5)	211/769 (27.4)	0.5485	(0.4549-0.6613)	<.0001
Random combined (13)	128/945 (13.5)	211/769 (27.4)	0.5288	(0.3717-0.7522)	.0004

removing the study in cancer patients by Abdelkefi et al.[77] A similar analysis for all catheter-associated deep vein thrombosis did not materially alter the summary relative risk; however, the estimate no longer reached statistical significance (relative risk, 0.42; 95% CI, 0.16 to 1.06). For fixed low-dose vitamin-K antagonist, the summary relative risk for symptomatic catheter-associated deep vein thrombosis increased from 0.60 to 1.28 (95% CI, 0.45 to 3.61)

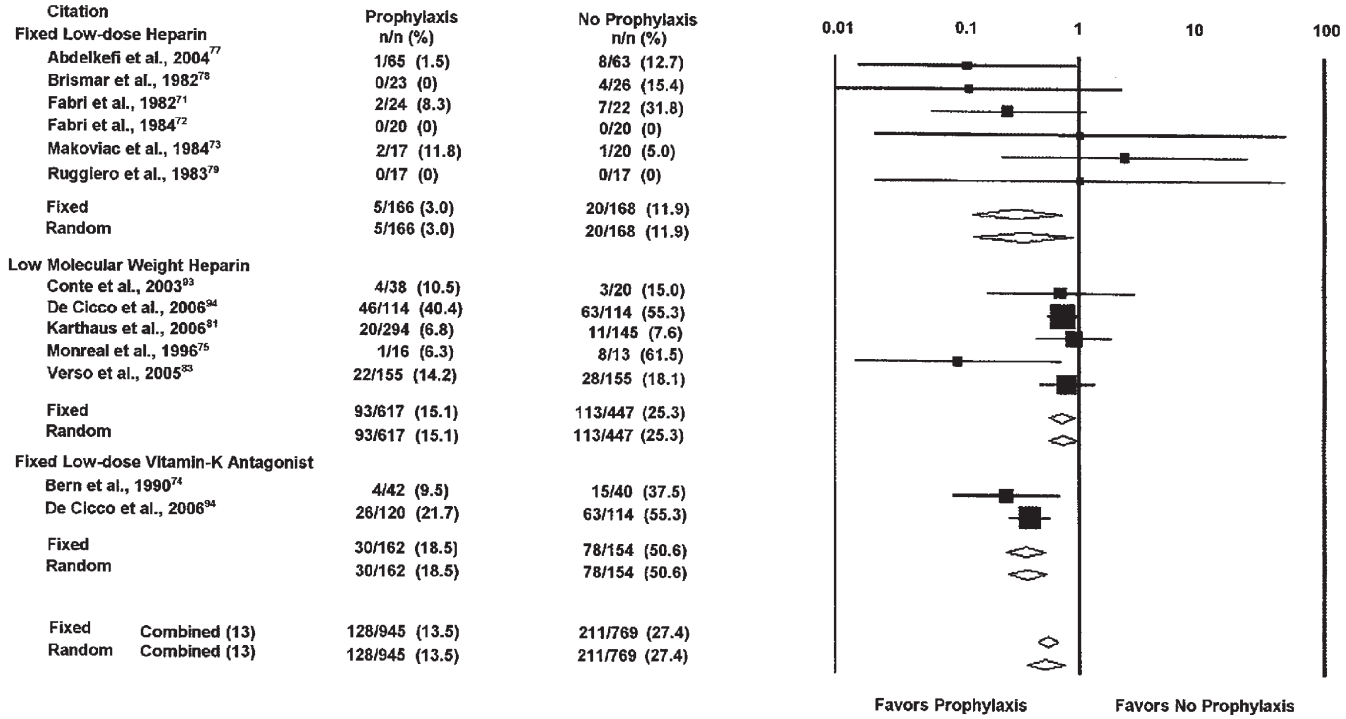

Figure 2 Relative risk of all catheter-associated deep vein thrombosis with anticoagulant prophylaxis.

Table 4 Relative Risk of Symptomatic Catheter-Associated Deep Vein Thrombosis with Anticoagulant Prophylaxis

Citation	Prophylaxis n/n (%)	No Prophylaxis n/n (%)	Relative Risk	(95% Confidence Interval)	*P* Value
Fixed low-dose heparin					
Abdelkefi et al, 2004[77]	1/65 (1.5)	5/68 (7.4)	0.2092	(0.0251-1.7430)	.1063
Fabri et al, 1984[72]	0/20 (0)	0/20 (0)	1.0000	(0.0208-48.0855)	1.0000
Fixed	1/85 (1.2)	5/88 (5.7)	0.2826	(0.0473-1.6886)	.1659
Random	1/85 (1.2)	5/88 (5.7)	0.3001	(0.0467-1.9269)	.2046
Low-molecular-weight heparin					
Karthaus et al, 2006[81]	10/293 (3.4)	5/145 (3.5)	0.9898	(0.3447-2.8423)	.9847
Monreal et al, 1996[75]	0/16 (0)	0/13 (0)	0.8235	(0.0174-38.9166)	.9213
Verso et al, 2005[83]	2/191 (1.1)	6/194 (3.1)	0.3386	(0.0692-1.6566)	.1595
Fixed	12/500 (2.4)	11/352 (3.1)	0.6890	(0.2995-1.5850)	.3808
Random	12/500 (2.4)	11/352 (3.1)	0.7177	(0.3047-1.6905)	.4480
Fixed low-dose vitamin K antagonist					
Bern et al, 1990[74]	4/42 (9.5)	13/40 (32.5)	0.2930	(0.1042-0.8238)	.0103
Couban et al, 2005[82]	6/130 (4.6)	5/125 (4.0)	1.1538	(0.3613-3.6849)	.8089
Heaton et al, 2002[76]	2/45 (4.4)	1/43 (2.3)	1.9111	(0.1798-20.3173)	.5840
Fixed	12/217 (5.5)	19/208 (9.1)	0.6039	(0.3044-1.1983)	.1492
Random	12/217 (5.5)	19/208 (9.1)	0.6039	(0.2250-2.1395)	.5247
Fixed combined (8)	25/802 (3.1)	35/648 (5.4)	0.5879	(0.3549-0.9740)	.0392
Random combined (8)	25/802 (3.1)	35/648 (5.4)	0.6113	(0.3584-1.0426)	.0708

after removing the study by Bern et al.[74] For low-molecular-weight heparin, the summary relative risk for symptomatic catheter-associated deep vein thrombosis increased from 0.69 to 0.98 (95% CI, 0.35 to 2.70) after removing the study by Verso et al.[83] No other summary estimates of relative risk were materially altered by removal of individual studies except that some estimates failed to reach statistical significance.

Limiting the analysis of all catheter-associated deep vein thrombosis to studies that were double-blind[73,81,83] increased the summary relative risk for low-dose unfractionated heparin infusion from 0.31 to 2.35 (95% CI, 0.23 to 23.75). For symptomatic catheter-associated deep vein thrombosis, limiting the analysis to studies that were double-blind[81-83] increased the summary relative risk for fixed low-dose vitamin-K antagonist from 0.60 to 1.15 (95% CI, 0.36 to 3.68) and increased the summary relative risk for any anticoagulant prophylaxis from 0.59 to 0.82 (95% CI, 0.41 to 1.62).

Limiting the analysis to studies published after the year 2000[77,81,83,93,94] did not materially alter any of the summary relative risks for all catheter-associated deep vein thrombosis. A similar analysis for symptomatic catheter-associated deep vein thrombosis[76,77,81-83] increased the summary relative risk using fixed low-dose vitamin-K antagonist from 0.60 to 1.28 (95% CI, 0.45 to 3.61). Further subgroup

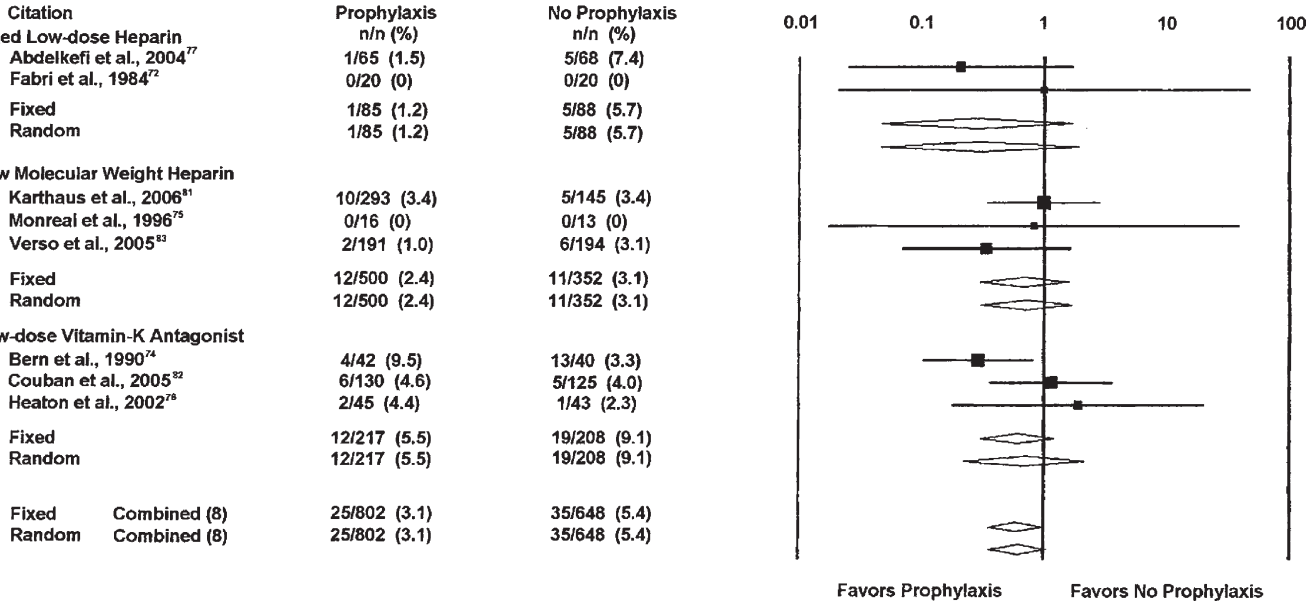

Figure 3 Relative risk of symptomatic catheter-associated deep vein thrombosis with anticoagulant prophylaxis.

Table 5 Relative Risk of Major Bleeding with Anticoagulant Prophylaxis

Citation	Prophylaxis n/n (%)	No Prophylaxis n/n (%)	Relative Risk	(95% Confidence Interval)	*P* Value
Fixed low-dose heparin					
Abdelkefi et al, 2004[77]	2/55 (3.6)	3/53 (5.7)	0.6424	(0.1118-3.6931)	.6168
Brismar et al, 1982[78]	0/23 (0)	0/26 (0)	1.1250	(0.0232-54.5411)	.9525
Fabri et al, 1984[72]	0/20 (0)	0/20 (0)	1.0000	(0.0208-48.0855)	1.0000
Fixed	2/98 (2.0)	3/99 (3.0)	0.7432	(0.1718-3.2157)	.6913
Random	2/98 (2.0)	3/99 (3.0)	0.7427	(0.1700-3.2445)	.6925
Low-molecular-weight heparin					
Conte et al, 2003[93]	0/38 (0)	0/20 (0)	0.5385	(0.0111-26.1756)	.7511
Karthaus et al, 2006[81]	1/285 (0.4)	0/140 (0.7)	0.4912	(0.0310-7.7957)	.6069
Monreal et al, 1996[75]	0/17 (0)	0/15 (0)	0.8889	(0.0187-42.2610)	.9523
Verso et al, 2005[83]	0/189 (0)	0/193 (0)	1.0211	(0.0204-51.1956)	.9917
Fixed	1/529 (0.2)	1/368 (0.3)	0.6582	(0.1177-3.6802)	.6339
Random	1/529 (0.2)	1/368 (0.3)	0.6526	(0.1143-3.7265)	.6312
Fixed low-dose vitamin K antagonist					
Couban et al, 2005[82]	0/130 (0)	3/125 (2.4)	0.1374	(0.0072-2.6333)	.1208
Heaton et al, 2002[76]	0/45 (0)	0/43 (0)	0.9565	(0.0194-47.1638)	.9822
Fixed	0/175 (0)	3/168 (1.8)	0.2400	(0.0270-2.1320)	.2003
Random	0/175 (0)	3/168 (1.8)	0.2788	(0.0265-2.9347)	.2875
Fixed combined (9)	3/802 (0.4)	7/635 (1.1)	0.5356	(0.2025-1.4163)	.2082
Random combined (9)	3/802 (0.4)	7/635 (1.1)	0.5923	(0.2146-1.6350)	.3121

analyses did not materially alter any other summary estimates of relative risk; however, some estimates no longer reached statistical significance.

The summary estimate of the relative risk of symptomatic catheter-associated deep vein thrombosis for any anticoagulant no longer achieved statistical significance using the more conservative random-effects model (relative risk, 0.61; 95% CI, 0.36 to 1.04). Using the random-effects model did not alter the results for any other outcomes.

Funnel plots of effect size versus precision show no evidence of publication bias. A similar number of studies were distributed on either side of the summary estimate, suggesting that studies with significant and nonsignificant results were equally represented in our analysis.

DISCUSSION

Our results indicate that all of the anticoagulant regimens evaluated are effective for preventing all catheter-associated deep vein thrombosis with relative risk reductions of 27% to 70% (absolute risk reduction 9% to 32%) (Table 3). The results also provide suggestive evidence that anticoagulant

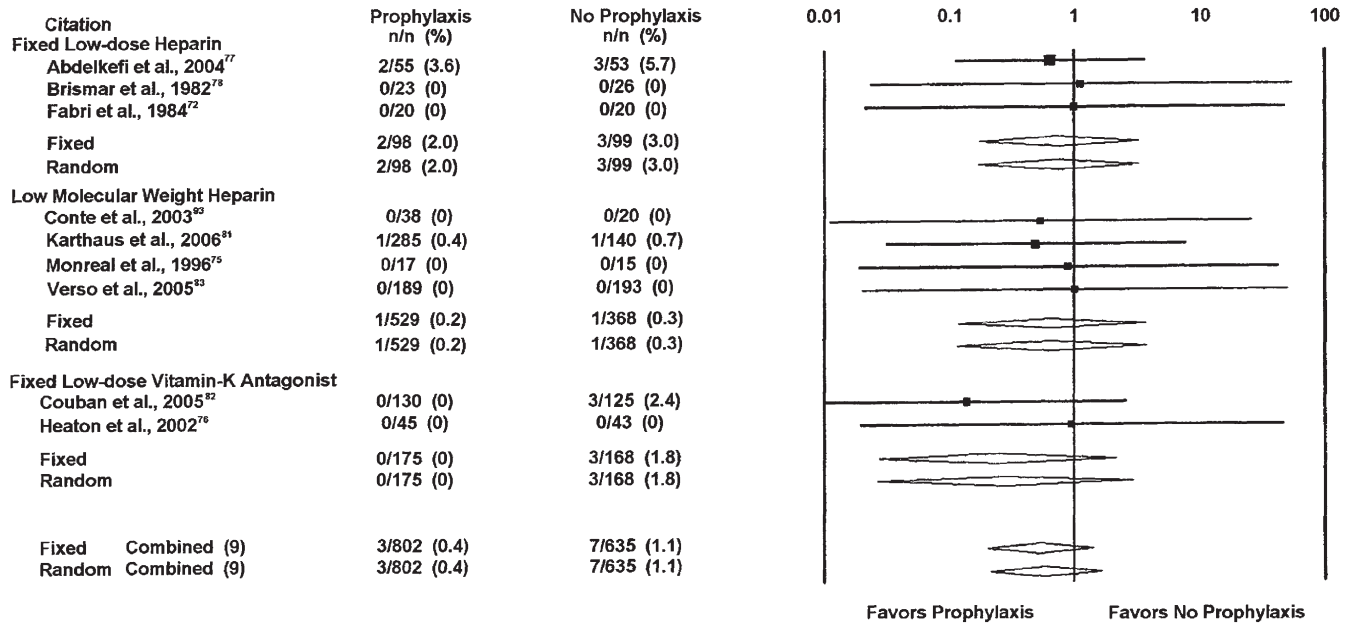

Figure 4 Relative risk of major bleeding with anticoagulant prophylaxis.

prophylaxis is effective for preventing *symptomatic* catheter-associated deep vein thrombosis, based on the pooled analysis across anticoagulant regimens (Table 4). The effectiveness of any individual regimen for preventing symptomatic catheter-associated deep vein thrombosis or pulmonary embolism remains uncertain due to the small number of symptomatic events in each subgroup. Confidence intervals for the summary relative risks were broad, however, and do not exclude the possibility of a potentially clinically important benefit.

The clinical importance of asymptomatic catheter-associated deep vein thrombosis is incompletely understood,[94,95] although catheter-associated deep vein thrombosis has been associated with clinically important pulmonary embolism,[2,3,96] including fatal embolism. Anticoagulant prophylaxis, therefore, has the potential to prevent serious thromboembolic events.

Our analysis does not support the view that the materials and structure of central venous catheters have improved recently, leading to lower rates of catheter-associated thrombosis. Limiting our analysis to studies published after the year 2000 did not change the reduction in relative risk of all catheter-associated deep vein thrombosis using any anticoagulant regimen, and the observed rates of all deep vein thrombosis without prophylaxis in these studies ranged from 7.6% to 55.3%. A similar analysis for symptomatic catheter-associated deep vein thrombosis increased the summary relative risk in only one subgroup of studies evaluating fixed low-dose vitamin-K antagonists. Due to smaller combined sample sizes, 95% confidence intervals for this estimate were broad and do not exclude the potential for a clinically meaningful benefit.

Our analysis found that the absolute risk of major bleeding was low for each of the regimens (Table 5). The pooled analysis did not detect an increase in major bleeding associated with anticoagulant prophylaxis. The absolute increase in the incidences of major bleeding with anticoagulant prophylaxis compared with no prophylaxis is unlikely to be greater than 0.9% for any regimen.

Analysis of 2 randomized trials[80,94] comparing fixed low-dose vitamin-K antagonist to low-molecular-weight heparin indicates that low-molecular-weight heparin is less effective for preventing all catheter-associated deep vein thrombosis than fixed low-dose vitamin-K antagonist prophylaxis (summary relative risk 1.9; 95% CI, 1.3 to 2.7). This finding is contradictory to results of recent clinical trials showing superior efficacy of low-molecular-weight heparin compared with vitamin-K antagonists for treatment of established venous thromboembolism in cancer patients,[97,98] and requires further evaluation.

We did not detect a statistically significant reduction in either pulmonary embolism or all-cause mortality with anticoagulant prophylaxis. Due to the low frequency of these outcomes, 95% confidence intervals for summary relative risks were broad and did not exclude the potential for meaningful benefit. Additional randomized controlled trials evaluating these outcomes are needed.

A recently completed randomized trial[89] compared fixed low-dose warfarin (1 mg) with either adjusted-dose warfarin (international normalized ratio 1.5 to 2.0) or no prophylaxis in cancer patients receiving chemotherapy via central venous catheters. Because the outcome data were reported for only 90% of patients, this trial was not included in our analysis. The results suggest that adjusted-dose warfarin was more effective than fixed-dose warfarin, reducing the incidence of symptomatic thrombosis from 7% to 3%; however, major bleeding increased from 2% to 4%. The observed rates of symptomatic thrombosis for fixed-dose warfarin or no prophylaxis were similar (5% and 6%, respectively, odds ratio 0.94; 95% CI, 0.52 to 1.72). The 95% confidence interval does not exclude the possibility of a clinically important benefit of fixed low-dose warfarin for preventing symptomatic thrombotic events. Because clinicians selected which patients entered into the randomized comparison of either fixed-dose warfarin versus no prophylaxis, or to the comparison of fixed versus adjusted-dose warfarin, the results may be biased against fixed low-dose warfarin if clinicians were effective at selecting patients at lower risk for thromboembolism for comparison with no prophylaxis, or patients at higher risk of thromboembolism for comparison against adjusted-dose warfarin.

Three previous systematic reviews have evaluated the efficacy of anticoagulant prophylaxis for prevention of catheter-associated deep vein thrombosis.[14,15,30] None of these reviews included 3 randomized controlled trials[77,81,94] that were included in our analysis.

Limitations of our meta-analysis include the potential for publication bias. Although our funnel plots did not suggest publication bias, the low number of trials makes definitive interpretation difficult. Our search strategy was comprehensive, including multiple databases, abstracts presented at national meetings, and manual searches for articles not retrieved by database searches. The overall limited number of trials retrieved, and their small sample sizes, limited our ability to precisely quantitate the efficacy of individual anticoagulant prophylaxis regimens for reducing symptomatic venous thromboembolism. In addition, tests of heterogeneity and sensitivity analyses were limited by the small number of trials. Finally, only 3 trials evaluating symptomatic outcomes were double-blinded with concealment of randomization.[80-82] These limitations all support the need for additional high-quality randomized controlled trials.

In conclusion, anticoagulant prophylaxis is effective for preventing all catheter-associated deep vein thrombosis. This finding is driven primarily by a reduction in asymptomatic thrombi detected by mandatory screening for deep vein thrombosis. The effectiveness of individual anticoagulant regimens for preventing symptomatic venous thromboembolism, including pulmonary embolism, remains uncertain. Based on the available evidence, the risk-benefit of anticoagulant prophylaxis for preventing clinically impor-

tant venous thromboembolic events is uncertain. With increasing central venous catheter use, there is a need for large rigorous randomized trials to resolve this issue. Currently, the decision to use anticoagulant prophylaxis remains a clinical judgment based on the estimated risks of thromboembolism and major bleeding in the individual patient.

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Table 1 Study Characteristics

Author/Year	Sample Size	Drug	Dose	Route	Frequency	Catheter Type	Vein Where Inserted	Duration Catheterization	Onset Prophylaxis	Duration Prophylaxis	Prophylaxis Continuous	Length of Follow-up
Fabri et al, 1982[71]	46	Heparin	3000 USP units/L	IV	With TPN	Central line	Subclavian	Not clear	After insertion	Not clear	Not Clear	Not Clear
Fabri et al, 1984[72]	40	Heparin	3000 units/L	IV	With TPN	Central line	Subclavian	22.1 ± 3.2 days	After insertion	Not clear	Yes	22.1 ± 3.2 days
Brismar et al, 1982[78]	49	Heparin	5000 U	IV	Every 6 hours	PICC	External jugular	7-94 days (mean 25 days)	At insertion	7-94 days (mean 25 days)	No	7-94 days (mean 25 days)
Ruggiero et al, 1983[79]	34	Heparin	1000 U/L	IV	With TPN	Central line	Subclavian	>14 days in 18 patients	Not clear	7-43 days (mean 18 days)	Yes	>14 days in 18 patients
Macoviak et al, 1984[73]	37	Heparin	1 unit/mL	IV	With TPN (2 L)	Central line	Subclavian	At least 4 weeks	After insertion	At least 4 weeks	No	4 Weeks
Abdelkefi et al, 2004[77]	108	Heparin	100 IU/Kg/D	IV	Continuous infusion	Central line	Subclavian	8-81 days	Not clear	8-81 days	Yes	8 Weeks after removal
Bern et al, 1990[74]	121	Warfarin	1 mg	PO	Daily	Tunneled	Subclavian	Not stated	Before insertion	90 days or DVT	No	90 days
Heaton et al, 2002[76]	98	Warfarin	1 mg	PO	Daily	Tunneled	Subclavian	30 Lines removed before 90 days	At insertion	Mean 41 days	No	90 days
Couban et al, 2005[82]	255	Warfarin	1 mg	PO	Daily	Tunneled, PICC, port	Subclavian, IJ, basilic, cephalic	Median 25 weeks	After insertion	Median 8 weeks	No	3 months after line removed
Monreal et al, 1996[75]	29	Dalteparin	2500 units	SC	Daily	Tunneled	Subclavian	Not stated	Before insertion	90 days or DVT	Yes	90 days
Conte et al, 2003[93]	58	Dalteparin	5000 IU	SC	Daily	Central line	Subclavian & IJ	>1 week	Not clear	7 days	Yes	7 days
Verso et al, 2005[83]	310	Enoxaparin	40 mg	SC	Daily	Tunneled, PICC	Subclavian, IJ and EJ	42 ± 2 days	Before insertion	42 ± 2 days	Yes	3 months
Karthaus et al, 2006[81]	425	Dalteparin	5000 IU	SC	Daily	Not clear	Not clear	At least 12 weeks	Within 5 days of insertion	16 weeks	Yes	16 weeks
Mismetti et al, 2003[80]	59	Nadroparin vs Warfarin	2850 units vs 1 mg	SC and PO	Daily	Tunneled	Subclavian and IJ	90 days except for 12 caths <90 days	Before insertion	90 ± 5 days or DVT	For first 90 days, yes	6 months
DeCicco et al, 2006[94]	450	Dalteparin vs Acenocumarine vs Control	5000 units vs 1 mg vs No Prophylaxis	SC and PO	Daily	Not clear	Not clear	Not clear	Before insertion	8 days	Not clear	Not clear

USP = United States Pharmacopeial Convention; L = liter; U = units; mL = milliliter; IU = international units; Kg = kilogram; D = day; mg = milligram; IV = intravenously; PO = orally; SC = subcutaneously; TNP = total parenteral nutrition; PICC = peripherally inserted central catheter; IJ = internal jugular; EJ = external jugular; DVT = deep venous thrombosis.

Table 2 Study Quality

Author/Year	Drug	Consecutive Enrollment	Randomization Concealed	Similar at Baseline	Treated Similarly	Patient Spectrum
Fabri et al, 1982[71]	Heparin	Not clear	Not clear	Yes	Not clear	Parenteral nutrition
Fabri et al, 1984[72]	Heparin	Yes	Not clear	Not clear	Not clear	Parenteral nutrition
Brismar et al, 1982[78]	Heparin	Not clear	Not clear	Yes	Not clear	Parenteral nutrition
Ruggiero et al, 1983[79]	Heparin	Yes	Not clear	Not clear	Not clear	Parenteral nutrition
Macoviak et al, 1984[73]	Heparin	Not clear	Yes	Yes	Not clear	Parenteral nutrition
Abdelkefi et al, 2004[77]	Heparin	Yes	Not clear	Yes	Yes	Hematologic cancer or hematologic disorders, age <60
Bern et al, 1990[74]	Warfarin	Yes	No	Not clear	Yes	Cancer
Heaton et al, 2002[76]	Warfarin	No	Not clear	Yes	No	Hematologic cancer
Couban et al, 2005[82]	Warfarin	Not clear	Yes	Yes	Yes	Cancer
Monreal et al, 1996[75]	Dalteparin	Yes	Not clear	Yes	Not clear	Cancer
Conte et al, 2003[93]	Dalteparin	Not clear	No	Not clear	Not clear	Hematologic cancer
Verso et al, 2005[83]	Enoxaparin	Yes	Yes	Yes	Yes	Cancer
Karthaus et al, 2006[81]	Dalteparin	Not clear	Yes	Not clear	Not clear	Cancer
Mismetti et al, 2003[80]	Nadroparin vs warfarin	Yes	Yes	No	Yes	Cancer
DeCicco et al, 2006[94]	Dalteparin vs Acenocumarine vs control	Yes	Not clear	Not clear	Not clear	Cancer

Table 2 Continued

Patients Blinded	Clinicians Blinded	Outcomes Assessors Blinded	Method Thrombus Detection	Major Bleeding Defined	Minor Bleeding Defined	All Patients Accounted For
Not clear	Not clear	Not clear	Venogram all patients	Not clear	Not clear	Yes
Not clear	Not clear	Not clear	Venogram all patients	Not clear	Not clear	Yes
No	No	Not clear	Venogram all patients	Not clear	Yes	Yes
Not clear	Not clear	Yes	Venogram all patients	Not clear	Not clear	Yes
Yes	Yes	Yes	Venogram all patients	Not clear	Not clear	Yes
Not clear	Not clear	Yes	Ultrasound all patients	Yes	No	Yes
No	No	Yes	Venogram all patients	Not clear	Not clear	Yes
No	No	Not clear	Venogram symptomatic patients	Not clear	Not clear	Yes
Yes	Yes	Yes	Ultrasound or venogram symptomatic	Yes	Yes	Yes
Not clear	Not clear	Yes	Venogram all patients	Not clear	Not clear	Yes
No	No	Yes	Echo-Doppler all patients	Not clear	Not clear	Yes
Yes	Yes	Yes	Venogram all patients	Yes	Yes	Yes
Yes	Yes	Not clear	Ultrasound and venogram all	Yes	No	Yes
No	No	Yes	Venogram all patients	Yes	No	Yes
Not clear	Not clear	Not clear	Venogram all patients	Not clear	Not clear	Not clear